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PROJECT PROPOSAL

LifeSign-AI: Touchless Emergency Assistance Station

1. Introduction

In India, emergency helplines such as 112 operate primarily through voice telephone calls. For speech and hearing-impaired people, this creates a serious safety problem during urgent situations like a sudden heart attack, kitchen fire, choking, or a physical assault. When an incident occurs, talking to an emergency operator is impossible, and typing an SMS or searching for a phone in panic takes far too much time. Common household devices do not solve this problem. Smart home voice speakers only respond to spoken words, standard security monitoring units only record video footage without understanding what is happening, and wearable emergency alert bands are usually taken off before sleeping or showering, leaving the person unprotected during the night.

Recent research has explored sign language detection and automated emergency alerts, but earlier approaches have clear practical drawbacks. Wearable glove systems [1] require users to keep wired, fragile hardware on their hands throughout the day. These gloves cause discomfort, break easily from repeated bending, and cannot be worn during sleep or bathing. Vision-based translation projects eliminate wearables [2, 3, 4], but they only work if the user stays directly in front of a stationary computer monitor or phone screen. Furthermore, standard optical sensors perform poorly in low lighting and completely fail to see gestures in dark rooms [5]. Systems that stream video feeds over the internet to external cloud servers [6] raise strong privacy concerns because users do not want private bedroom or living room video sent to outside companies, while also causing delays and failing if the internet goes down. In addition, existing automated alert prototypes [7] have no false-alarm cancel option, meaning ordinary everyday arm movements can mistakenly summon emergency services. Many of these systems also lack a dedicated app for family members, sending only a single unverified SMS, and estimate user location through internet routing, which can be off by several kilometers instead of giving the exact house address.

To solve these issues, we proposed LifeSign-AI as a self-contained, touchless emergency station for the home. The system uses a motorized vision sensor that rotates to follow the person as they move around the room. It has built-in night vision, allowing it to recognize emergency hand gestures even in complete darkness. To protect user privacy, all video processing is performed locally inside the room on an offline computer, so no personal footage is ever uploaded to the cloud or shared over the internet. To prevent false alarms caused by casual hand gestures, the system includes a physical abort button along with an easy cancel gesture, allowing accidental triggers to be stopped immediately before help is dispatched. Once an emergency is confirmed, the station sounds a loud in-home buzzer and turns on a flashing strobe light to alert neighbors. At the same time, it automatically sends emergency text messages with exact satellite GPS coordinates to emergency helplines and family contacts, while pushing instant alerts to a dedicated companion mobile and web application. This companion app provides family members and caregivers with immediate notification of the specific emergency type (such as medical, fire, or choking), displays live map directions to the home, and shows the real-time status of the safety station. A physical push button is also included as a backup in case the optical sensor is blocked or turned off.

2. Project Objectives

The primary objective of the proposed system is to design a smart emergency alert system that interprets some Indian sign language (ISL) gestures and can identify emergency situations, by analyzing the movement of the hands over a certain period of time and confirming the gestures before any alerts are triggered to reduce false alarms. The proposed system will also provide a demonstration of the camera's infrared night vision for emergency gesture detection during nighttime or in dark conditions.

The secondary objective is to introduce ways of increasing the reliability and accessibility of the emergency alert system using GPS location sharing, SMS alerts, and physical emergency button with cancellation from an ESP32 platform-based system. The system introduces a safety-based alert process, local video processing for privacy, and failure-handling mechanisms for camera, landmark, and network issues. The proposed system has a built-in physical button to raise and cancel distress signals and provide manual and automatic approaches to requesting assistance.

The mobile application provides the interface for monitoring the system and receiving emergency information. The proposed system provides with both automatic gesture-based and manual methods for requesting assistance.

3. Literature Survey Summary (Limitations of Prior Work vs. LifeSign-AI)

SN o	Paper Title	Methodology used	Performance metrics	Limitations
1	AI-Enabled Sign Language Recognition and Communication System For Speech/Hearing-Impaired Users IEEE International Conference on Smart Internet of Things (SmartIoT) 2024	- triboelectric sensing glove (15 sensors) capturing hand motion signals - 1D Convolutional Neural Network (CNN) for non-segmentation word/sentence recognition - segmentation-assisted CNN that splits sentence signals into word fragments for recognizing new/unseen sentences - bidirectional Virtual Reality (VR) interface for two-way signer/non-signer communication	- 20 isolated words: 93.25% accuracy - 10 continuous sentences: 95.5% accuracy - Segmentation-based fragment recognition: ~82% (81.9%) accuracy - New/never-seen sentence recognition: 86.67% average correct rate	- requires wearing a wired sensor glove continuously, which is less convenient than a camera-only, glove-free system - flex/triboelectric sensor wiring is delicate and prone to mechanical wear - evaluated only on a small, fixed vocabulary (20 words / 10 sentences); scaling to open vocabulary is untested
2	Sign Language Interpretation to Text: A Real-Time CNN-Based Approach IEEE International Conference on ICT in Business Industry & Government (ICTBIG) 2025	- webcam-based image acquisition with cvzone hand detection and ROI extraction - preprocessing via flipping, grayscale conversion, and resizing to 400x400 - Convolutional Neural Network (CNN) trained on the ASL Alphabet dataset for static gesture classification - post-processing with dictionary-based word correction (enchant) and	- Overall classification accuracy: 98% - Precision: 98%, Recall: 97.5%, F1-score: 97.7% - Real-time inference latency: under 100 ms per frame	- handles only static gestures; dynamic letters such as "J" and "Z" require temporal models (RNN/LSTM/3D CNN) - system recognizes individual alphabet letters only, not continuous words or sentence-level signs - no emergency alert, notification, or dispatch

		text-to-speech (pyttsx3) output		mechanism is included
3	Sentence-Level Sign Language Recognition Using MediaPipe RIVF International Conference on Computing and Communication Technologies (RIVF) 2024	- MediaPipe Hand and Pose used to extract 46 skeletal landmarks per frame from recorded sign videos - landmark normalization (wrist-relative) and mean-filter interpolation to handle missing landmarks - novel TDCNN + CNN hybrid (Time-Distributed CNN for spatial features, CNN for temporal features) for word-level recognition - multi-window sliding filter (0.4s-0.8s) aggregated by averaging to assemble sentence-level predictions from word-level detections	- Word-level accuracy: 99.42% (custom dataset), 98.5% (WLASL100), 96.93% (AUTSL), 96.65% (WLASL300) - Sentence-level evaluation: BLEU score of 0.883 and Word Error Rate (WER) of 0.081 using a 4-window filter	- 2D skeleton data causes occlusion issues, giving identical results for crossed/overlapping key points - each additional detection window adds ~15 ms processing time, creating a trade-off between accuracy and real-time speed - evaluated only on curated/controlled datasets, without a real-world emergency-alert use case
4	Sign Language Converter For Deaf and Mute People IEEE International Conference on Artificial Intelligence and Machine Learning Applications (AIMLA) - Theme: Healthcare and Internet of Things 2026	- standard webcam captures hand gestures in real time; OpenCV used for background removal and hand-region extraction - Convolutional Neural Network (CNN) trained on labeled Indian/Gujarati Sign Language alphabet datasets - language-mapping module converts predicted English alphabet output to Gujarati Unicode text - GUI-based Text-to-Actions module maps recognized/typed text to corresponding gesture animations	- Recognition precision: over 92% for clearly performed signs in well-lit environments - Real-time frame rate: 18-22 FPS	- prediction accuracy drops with partially obscured gestures and in low-light conditions - system currently recognizes only individual letters, not full words or sentences - requires the user to remain within camera view; no tracking, alerting, or dispatch functionality
5	Sign Language Recognition and Translation Systems for Enhanced Communication for the Hearing Impaired IEEE International Conference on Cognitive, Green and Ubiquitous Computing (IC-CGU) 2024	- ROBITA Indian Sign Language Gesture Dataset and a custom English alphabet/digit dataset used for training - Gaussian, Median, and Bilateral filtering applied for image preprocessing and background removal - "Reversible CNN" (12-layer) architecture that both recognizes gestures and reconstructs the original input for sign-to-text conversion - separate Text-to-Sign module using NLP-based concept matching and 3D-avatar/motion-capture animation	- Sign-to-text conversion overall accuracy: 70% - Precision: 46.6428%, Recall: 75.5314%, F1-measure: 49.4197% - Kappa coefficient (inter-rater reliability): 40.3191%	- limited dataset size (only 23 isolated ISL gestures) restricts generalizability and prediction accuracy - potential demographic and data-collection biases due to a controlled recording environment - relatively low precision/F1-measure compared to other CNN-based systems, indicating unreliable real-world performance
6	Sign Language Recognition and Translation to Text and Audio for Divyang Communication as a Microservice IEEE International Conference on Electronics and Computing, Communication Networking Automation Technologies (ICEC2NT) 2025 (Sep 3-4, 2025)	- hybrid microservice architecture with an Input Module (OpenCV, MediaPipe, YOLO for gesture capture), Processing Services (NLP-based gloss-to-text via Transformer models), and a Data Management layer - CNN-based gesture classification mapped to sign glosses, then converted into natural-language text using Seq2Seq/Transformer NLP models - Text-to-Speech synthesis (gTTS/Amazon Polly/Azure/Tacotron/WaveNet) for audio output - API Gateway (Spring Boot/Express.js) for secure, scalable cross-platform integration	- Reported sign-to-text conversion accuracy: 88%	- system architecture depends on networked microservices and cloud-hosted databases, introducing latency and internet-dependency risk - continuous video streaming to backend services raises data-privacy concerns - no dedicated emergency detection, false-alarm cancellation, or location-based alerting mechanism
7	With You - Indian Sign Language Detection and Alert System	- real-time ISL emergency recognition using MediaPipe Holistic to extract 3D	- Alphabet signs: 87.50% accuracy	- no cancellation window or abort button is described; the

IEEE International Conference on Emerging Smart Computing and Informatics (ESCI) 2024 (Mar 5-7, 2024)	landmarks (30 frames, 126 coordinates) - sequential LSTM network for classifying emergency gestures - deployed on a web app ("With You - Emergency Buddy") - triggers automated SMS using the Fast2SMS API with emergency type and IP-based location	- Dynamic signs: 97.20% accuracy - Emergency signs (6 classes): 88.88% accuracy	paper does not address false-alarm recovery - location is shared only as "IP address and geographical location" in the alert message; the paper does not specify or evaluate GPS-level location accuracy - single-channel SMS alert only, with no companion app or automated voice-call fallback described
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LifeSign AI tries to get around the big limits in other emergency systems. It spots distress through hand gestures on camera, so no gloves or sensors are needed at all. The setup tracks the whole room all around and adds night vision, which cuts blind spots and still picks up gestures when the light is low. Everything runs locally and offline, so video does not leave the home. That seems to lower privacy issues and any network delays that might happen.

Dual trigger activation uses gesture recognition and a physical button to make sure that the responses are more reliable. To stop the false alarms, cancel options with a button and a gesture are also used. When an emergency is confirmed, the system turns on an siren and sends SMS alerts with GPS location to pre-defined contacts. A mobile app checks family details and status from anywhere.

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1.PROPOSED WORK WITH METHODOLOGY

The Proposed architecture is shown in Figure 1.

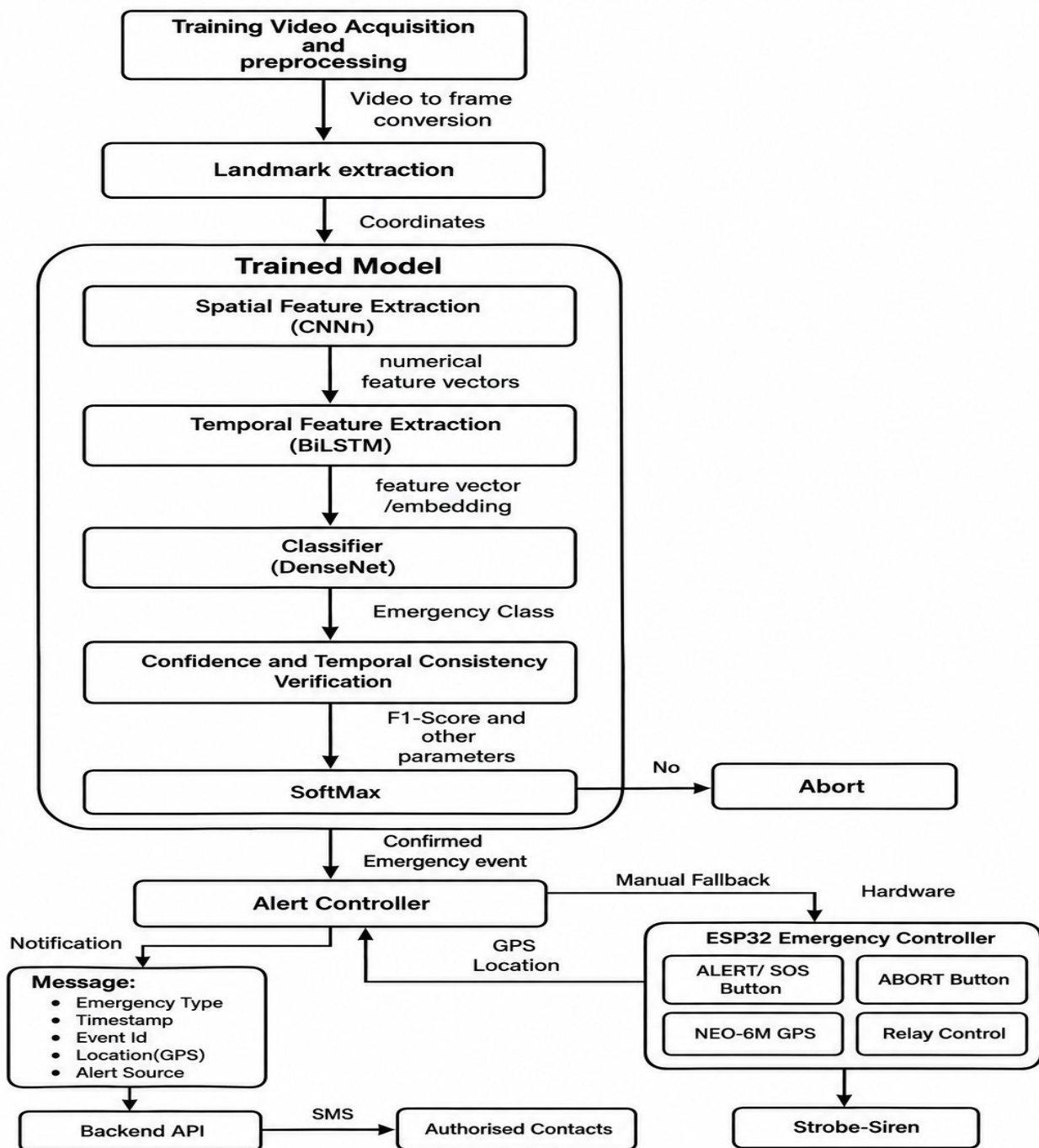


figure 1 architecture

The module of the architecture is explained below.

1. Video Acquisition and preprocessing

The Tapo c200 is primarily used to capture the user's live video. It then transmits the video stream over a wifi network using RTSP(Real Time Streaming Protocol). The host receives the video stream and converts the video into frames. The Tapo provides 1080p video , PTZ (Pan,Tilt, Zoom) movement , 360° pan and 114° tilt, motion tracking , IR night vision and Local RTSP and ONVIF support. The system does not depend on external servo motors for Tapo Movement .The openCV handles the frames and converts them into a format suitable for the landmark extraction model.

2. Landmark Extraction

The preprocessed frames are passed to the MediaPipe Holistic, which extracts the landmarks in hands (also body and head). Each hand has 21 landmarks and these are given as coordinates . MediaPipe Holistic additionally provide information about body/pose which can be incorporated to provide contextual body information. Frames with partially detected or missing landmarks are handled using landmark validity checks.

3. Spatial Feature Extraction and Normalisation

The extracted landmarks are converted into normalized numerical feature vectors. The spatial relationships among the extracted landmark features are learned using a one-dimensional Convolutional Neural Network (Conv1D). This learns the relationships between wrist, elbow, shoulder and other points , this helps in reducing the sensitivity to the user's position , scale and distance from the camera. Additionally it computes the relative distance , velocity and acceleration to make the system more motion aware. This helps to distinguish gestures not only based on hand position but also on how it moves over time. Normalisation ensures that the model is not sensitive to the user's distance from the camera , the hand position in the frame and the difference in the hand size between the users.

4. Temporal Feature Extraction

Sign Languages are dynamic and cannot be identified accurately using a single frame. Thus , the system stores a sequence of consecutive frames in a temporal window using Bi-Directional Long Short Term Memory (BiLSTM) The training data contains gestures

performed at different speeds so that the model becomes less dependent on a particular signing speed. It learns the temporal dependencies and movement patterns of the gesture before the final classifier predicts the emergency class. This helps the system to distinguish the gestures that may have similar individual frames but different movement patterns.

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7. Emergency Gesture Classification

The temporal representations are passed to the fully connected Dense layer. It learns the temporal dependencies and movement patterns of the gesture before the final classifier predicts the emergency class . Here we also use a Softmax function to estimate the probability of the predefined class.

The emergency gestures are classified into the following classes

- HELP / SOS
- FIRE
- MEDICAL EMERGENCY

- ACCIDENT / CRASH
- CHOKING
- CANCEL
- IDLE / NON-EMERGENCY
- UNKNOWN

The class with highest probability is considered as the model's predicted gesture.

8. Confidence and Temporal Consistency Verification

The single classification result is not immediately considered as an emergency. It does additional verification that includes

- Temporal Consistency (persistence across the consecutive temporal window)
- Gesture Duration
- Prediction Confidence Score
- Emergency Class Validity

This prevents false alarms due to incorrect frame or random movement.

9.Alert Controller

After verification of the emergency gesture and its class, the system creates an emergency message in the backend containing the following details, which are then sent as an alert to the nearby authorised person:

- Emergency type
- Timestamp
- Event ID
- Location (GPS)
- Alert source

An authorized SMS notification mechanism can then notify the predefined emergency contacts with the emergency details and location.The Alert Controller also helps maintain a controlled alert lifecycle and prevents unverified or repeated predictions from directly generating emergency notifications.

10. ESP32-Based Manual Emergency and Abort Mechanism

To provide an independent fallback mechanism, an ESP32 is integrated with physical ALERT/SOS and ABORT/CANCEL buttons.

The AI-generated emergency request is passed to the same Alert Controller.

The ESP32 can also control a relay connected to a local siren or strobe indicator, providing a physical emergency indication even if the AI recognition process fails. It communicates with the host computer through USB serial communication .

11.Location Acquisition

When an emergency is confirmed, the system obtains the user's location using the integrated location mechanism. The proposed hardware configuration uses a NEO-6M GPS module connected to the ESP32.

The location information is attached to the emergency event before notification. If location information is unavailable, the system explicitly reports the location as unavailable rather than generating an incorrect location.

1. IMPLEMENTATION

Tools and Languages Used: Python, C/C++, ESP32, JavaScript, React

Libraries and Frameworks: OpenCV, MediaPipe Holistic, NumPy, PyTorch, TensorFlow/TFLite, FastAPI/Flask, Node.js/Express, PostgreSQL, Arduino Framework, RTSP

Implementation Summary:

This system recognises emergency related Indian Sign Language(ISL) and provides alerts along with the location.

1. The live video stream is retrieved by the Tapo C200 and then is sent to the host which converts the video stream to frames.
2. The MediaPipe Holistic extracts the landmarks from the incoming frames and handles missing or partially detected landmarks.
3. Gestures are collected and are categorised in the various emergency and non-emergency classes. Here we generate normalised landmarks and compute motion related features like relative position, velocity, acceleration etc..
4. For the AI model development, the Conv-1d for the spatial features extraction and the BiLSTM for the temporal features extractions is used. These are then classified into different emergency classes by Dense classifier and SoftMax. Evaluate model using accuracy, F1 score and confusion matrix.
5. Additional verifications are carried which include temporal consistency checks.
6. The ESP32 provides alert and abort buttons. The relay module is connected to control the local strobe/siren. USB serial communication is established between the ESP32 and host. This is also interfaced with the NEO-6M GPS Module.
7. The SMS is then sent to the verified contacts, this includes emergency types, timestamp, GPS location etc..
8. Flutter is used to create a mobile app, providing users with an interface. The trained AI model is deployed with the app to support real-time gesture recognition and emergency.

2. WORK PLAN

PHASE	WORK	DELIVARABLE
1.	Hardware setup and finalization of ISL signs to be recognized by a camera.	Circuit Design.
2.	Record the ISL Signs from different people and collect the movements through the RTSP video feed.	Labelled ISL landmark dataset collection.
3.	Develop the OpenCV and MediaPipe pipeline to process the video and extract the movement points and prepare the data for recognition.	Real-Time feature processing pipeline.
4.	Train the conv1D-BiLSTM model and implement the LSTM and GRU models for the comparison.	Trained models.
5.	Evaluate the Temporal Transformer model if the available data and computing resources permits.	Model Comparison.
6.	Implement confidence checks, persistence, cooldown, and state-based decision logic to reduce the false alerts.	Safety checking system.
7.	Integrate the ESP32 ALERT/ABORT with the relay and siren.	Automatic and Manual Mode Prototype.
8.	Integrate the GPS location, notifications (Alert message with GSM backup) and dashboard.	Complete Working System.
9.	Test the system during Wi-Fi loss, power loss and other possible failures. Fix the problems and perform the final testing.	Final Validated System.

3. EXPECTED OUTCOME/RESULTS

Since we have developed the projects in different modules, each stage of the project will provide a useful output.

1. The Expected Outcome is a working prototype that recognizes a small number of ISL signs from live video and when the system correctly recognizes the emergency sign, it will start the SOS alert process.
2. The system also supports the hands-free AI activation and ESP32 ALERT button as a backup, ABORT/CANCEL protection, GPS location , notification status and event tracking.
3. Instead of using external gimbals, we use the TAPO C200 camera which has a built-in 360° view and 114° vertical view with autotracking. It also removes the limitation of mechanical gimbal's movement range of limited mechanical tilt range.
4. The hardware guide gives a target value of more than 96.5% validation accuracy, about 11.4ms model inference time and less than 80ms local RTSP delay from TAPO C200. These are the only target values from the source document. They are not Guaranteed project results.
5. The final report will show the actual result which is obtained from the collected dataset. It will check how the system detects emergency signs and it also checks how many false alerts it gives, instead of checking only the overall accuracy.
6. The prototype dataset will have a starting target of 100 samples per class for 8 classes , giving the total of 800 sequences. Each sequence will have 30 frames and 126 features per frame. Multiple signers will be recorded with different lighting, background, distance, angle, speed, occlusion and normal movements.
7. The system will be tested using signer-independent data and different conditions such as lighting, distance, camera failure, network failure, GPS failure and repeated activation. The final report will include accuracy, precision, false-alert rate, emergency recall, inference FPS , notification-acknowledgment latency, model size, missed emergency rate.

Further improvements and research will be done to improve emergency sign detection , reduce false alerts and missed emergency rate and make the system faster and more reliable for real world emergency situations.

9 .APPLICATIONS

1. This system can be used to provide emergency help for Deaf and speech-impaired people in homes, colleges and workplaces and it can also be useful in hospitals and care centres to help send emergency alerts to trusted contacts and it can also be used in assisted-living and care environments, where a single camera can reduce the complexity of installation.
2. This system also uses the Tapo C200 ceiling- mount option to provide wider room coverage in smart rooms and kiosks and in the future, the system can be deployed on edge devices using compact landmark models, while for the local video storage, the camera's MicroSD recording will be used and the camera can monitor the user even in dark or low-light conditions using infrared night vision.
3. If the AI system fails to recognize the sign correctly, a physical SOS button can be used to send an emergency alert and this system can be used in residential rooms to monitor users who may be alone during emergency situations and this system also alerts nearby people through the local sirens and visual indicators, which enables immediate assistance during an emergency.
4. This system can be used to send emergency SMS alerts which contains user exact location, helping family members or emergency responders quickly identify and reach the user and this system also used to prevent from accidental emergency alerts by allowing user to cancel a false alarm through a physical button before the alert is dispatched.

CONCLUSION

This project is proposed as a safety-oriented emergency communication system rather than a simple sign-language recognition system. The main aim is to help a person communicate in an emergency situation when they are unable to communicate through speech.

The system uses a camera to observe the person's hand and body movements, and it identifies the emergency signs made by the person. To reduce the possibilities of false alerts, the system checks the recognized sign before sending an emergency alert.

We are using Tapo C200, it reduces the requirement for extra wiring and mechanical parts, which can also reduce the chances of hardware failure. This camera also provides night vision capabilities, automatic movement tracking and wide viewing area.

The system also handles situations, when the Wi-fi or Internet connection is not available during an emergency, It includes the backup method to improve the chances of delivering alerts and the system also avoids sending the same alert repeatedly and it checks the user's location when required. The user information is handled within the system to improve safety and privacy, whenever possible.

Overall, the project aims to provide a more reliable emergency alert system by considering the problems such as incorrect recognition, network failure, false alerts and hardware failures, the proposed system is designed to provide better support during real emergency situations.